LP, SDP and Quasi-convex Reformulations

Convex Optimization, Homework 4

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Problems and what I did

1. **Norm problems as LPs.** I rewrote $\max \|Ax-b\|_1$ s.t. $\|x\|_\infty \le 1$ (and the swapped version) as LPs. The vector $u \ge |Ax-b|$ and the scalar $t \ge |x_j|$ make every constraint linear, and at the optimum $t = \|x\|_\infty$ and $u = |Ax-b|$.
2. **Matrix-fractional problems as SDPs.** With an epigraph variable t and the Schur complement, constraints of the form $c^T F(x)^{-1} c \le t$ (with $F(x) \succ 0$) become LMIs in (x, t). A second variant was handled through duality and a slack variable.
3. **Quasi-convex problems.** A ratio of a convex numerator to a concave positive denominator is quasi-convex. I wrote the epigraph form and noted that it can be solved by bisection. A variant with a squared ℓ_1 term can be written as a QCQP or an SOCP.
4. **Lamp illumination.** Total illumination is a pointwise maximum of affine functions, so it is convex. The set where at least half the patches are dark is a superlevel set of a concave function, so it is convex. The number of lit patches is a sum of step functions, so it is not convex.
5. **A trace minimisation.** With $A \succ 0$ the constraint is equivalent through the Schur complement to a PSD condition, and the matrix minimising $\operatorname{tr} X$ is the smallest one in the PSD order. A related piecewise-linear objective was rewritten as an LP with auxiliary variables v, w.
6. **Separating two point clouds by an LP (`Q8.py`).** The constraints are $w^T a_i + b \ge 1$ and $w^T b_j + b \le -1$, and the objective is $\min \|w\|_1$. I generated the two clouds myself (20 Gaussian points around $(2,2)$ and 20 around $(-2,-2)$), solved the LP with CVXPY and plotted the separating line.
7. **Distance between two polytopes (`Q9.py`).** A QP: $\min \|x-y\|_2^2$ s.t. $A_1 x \le b_1$, $A_2 y \le b_2$, tested on a unit square and a shifted square.
8. **Optimal set point of a CSTR (`Q10.m`).** The steady state satisfies $(I-A)x = Bu$, with bounds $|u| \le 15$, $|x_1| \le 0.5$, $|x_2| \le 5$. The cost is piecewise quadratic in x_2 over three regions. I modelled the regions with binary variables and big-M constraints, and solved the mixed-integer QP with YALMIP's branch-and-bound. Re-running the script with YALMIP gives $x_1 = -0.0358$, $x_2 = 2.0011$, $u = -2.6116$, cost -0.0358: the optimum sits at the edge of the zero-cost region $|x_2| \le 2$.

Note: `Q8.py` and `Q9.py` need CVXPY, which could not be installed on the machine used to prepare this repository, so they were not re-run here.